

QIWEN(KYRA) JIAO

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EDUCATION

Master of Engineering, Electrical and Computer Engineering Sep. 2025 – Dec. 2026
University of Waterloo, Waterloo, ON, CA (GPA: 3.9/4.0)

Bachelor of Applied Science, Honours Computer Science (Co-op) Sep. 2020 – Jun. 2025
McMaster University, Hamilton, ON, CA (GPA: 3.7/4.0)

EXPERIENCES

Software Development Engineer (Intern) May 2026 - Aug 2026
Quantumera AI, Remote (U.S.)

- Built and deployed an AWS-based transportation data ingestion pipeline using SQS and EC2, integrating DynamoDB data and storing processed outputs in S3 with IAM-based access control.
- Created an ElastiCache Redis cluster to cache frequently accessed data, reducing end-to-end latency by 33%.
- Processed S3 data with EMR and Spark and indexed outputs in Elasticsearch to support downstream analytics and AI models.

Software Development Engineer (Intern) Aug. 2023 – Aug. 2024
Bosch (China) Investment Ltd, Suzhou, China

- Independently developed and deployed an enterprise-level web system integrating frontend and backend microservices on distributed servers, attracting 5k+ active users and handling 1M+ user requests in parallel.
- Designed and implemented reusable frontend components with Angular/Vue, improving UI consistency across 20+ modules. Utilized Pinia, RxJS, and Redux for state management and efficient data flow, reducing redundant logic by 30%.
- Developed backend services with Spring Boot and MySQL, implementing optimized queries and RESTful APIs documented via Swagger UI. Improved response time by 30% and reduced integration defects by 25%.
- Integrated CI/CD pipelines and Dockerized deployments using GitHub Actions under Linux environments, reducing release time by 40% and ensuring distributed reliability.
- Collaborated cross-functionally to design and test APIs, integrating text parsing and data normalization modules to standardize input formats and improve data consistency across services; achieved a 99% on-time delivery rate following Agile practices.

Automation Test Engineer (Intern) Jun. 2021 – Dec. 2021
Bosch (China) Investment Ltd, Suzhou, China

- Developed and maintained automated test suites using JUnit, designing unit, integration, and regression testing to ensure system reliability and performance.
- Designed detailed test plans and scenarios based on product requirements, executing 200+ test cases, and identifying 50+ critical defects. Worked closely with developers to provide actionable feedback and improve overall product quality.
- Integrated automated tests into CI/CD pipelines with GitHub Actions, enabling continuous validation during builds. This reduced release cycle time by 30%, improved early defect detection.

PROJECTS

Hammerly, Online Bidding Platform with AI customer support Jan. 2026 – Aug. 2026
React, TypeScript, JavaSpring Boot, Kafka, Redis, PostgreSQL, RAG, LLM APIs, Docker, Kubernetes, GCP, Prometheus, k6

- Led development of a full-stack online bidding platform with 5+ core workflows and 15+ RESTful APIs, using React, Spring Boot, Spring Security, JWT authentication, PostgreSQL, and Redis for auction, bidding, account, and session data.
- Built a RAG-based AI customer support microservice integrating LLM APIs, embeddings, and vector retrieval, and introduced Kafka-based asynchronous processing to decouple user-facing APIs from latency-intensive AI workloads and absorb burst traffic.
- Containerized and deployed services on Kubernetes in GCP with horizontal scaling; conducted k6 load testing for 1,000+ concurrent users and monitored throughput, P95 latency, error rates, JVM metrics, Kafka lag, and service health with Prometheus to identify and optimize performance bottlenecks.

TECHNICAL SKILLS

Languages:	Python, Java, JavaScript, TypeScript, SQL
Backend & Frontend:	Spring Boot, Django, Node.js, React, Angular, Vue, REST APIs
AI & Data:	Generative AI, LLM APIs, RAG, NLP, Apache Spark, Elasticsearch, Redis
Cloud & DevOps:	AWS, GCP, Docker, Kubernetes, GitHub Actions, Linux